

9. Discussion, Limitations and Recommendations

9.1 - Discussion, Limitations and Recommendations

The local validation establishes that the synthetic January 2025 data can pass the implemented processing and reporting path with reconciled row counts and billed cost. It also exposes a limitation of the analytical product: 95.98% of billed cost in that month is assigned to an `Unknown` cost center (Section 6.3). A reconciled total therefore does not, by itself, provide usable accountability. The allocation rules and source attributes need business review before the same measures can support managerial showback.

The project implements a versioned contract, a canonical Silver representation, a Gold model and subject-oriented datamarts. The independent generator makes the input reproducible. These properties support the reliability question in section 1.3 for the local tests. The Belgium plan reports a historical Databricks backfill, but its run identifiers and control outputs are not yet part of the thesis evidence package. A complete Daily promotion, monthly close and recovery test would address claims that the current local results cannot establish.

The governed self-service question remains less resolved. The local Streamlit application executes against datamarts, but the intended manager and analyst identities have not been tested across dashboard, export and direct SQL access. The existing Power BI dashboard already serves several manager groups according to the student's account. A future user study must therefore compare defined tasks and change requests, rather than assume that introducing another interface creates autonomy. Security and business approval remain conditions for any wider access.

Cost comparisons in chapter 6 are descriptive results from generated data. They do not show avoidable expenditure or savings at the host organization. Similarly, chapter 8 defines the platform cost boundary without a complete DBU, GCP and labor-cost dataset. The first economic recommendation is to record comparable workload costs and outputs; only then can Serverless, Classic, scheduling or retention changes be ranked. Performance tests must control input, revision, cache and compute configuration as set out in section 7.3.

The evidence has limited external validity because the detailed history is synthetic and the validated local dashboard covers one month. The construction demonstrates that the engineering method is testable, while actual billing semantics, utilization, user tasks and organizational benefit remain unmeasured. The next validation priority is to capture cloud run artifacts and access tests, then evaluate representative business scenarios on authorized data.

Centralized production and domain use

The proposed model keeps shared ingestion, definitions and controls with the platform and FinOps functions while distributing access to approved products. This can support independent consumption without assigning production responsibility to each domain. The project has not tested whether such an arrangement shortens change requests. Section 9.2 examines an embedded Data Engineer as one possible response to the reported handoffs; its staffing cost and effect require measurement. The local single-domain prototype gives no evidence for adopting a full Data Mesh operating model.

9.2 - Potential Contribution of a Data Engineer Embedded in FinOps

9.2.1 From observation to hypothesis

The workflow in section 1.2 separates management feedback, relay by the manager and implementation by an external contributor. The issue extends beyond dashboard access to FinOps' ability to change data, indicators and reporting directly.

A Data Engineer close to the FinOps business team could bring requirement interpretation nearer to implementation. This proposal does not remove supervisor approval or automatically replace the Data Platform and Power BI teams.

9.2.2 Capabilities and potential benefits to assess

Dimension	Potential contribution	Required evidence or condition
Responsiveness	More direct translation of business feedback into a testable change	Compare end-to-end delay, waiting and iterations
Data handling	Adapt pipelines, contracts, datamarts and aggregates within FinOps	Authorized skills, permissions and technical scope
Business consistency	Validate cost definitions and investigate inconsistent KPIs with FinOps	Documented business reviews, tests and reconciliation
Knowledge retention	Retain context, decisions and knowledge within the team	Documentation, version control, code review and continuity
Human cost	Possible reduction in coordination time or some external work	Total-cost comparison at equivalent scope and quality, including the embedded role

Changing visuals, DAX measures or Power BI models may require specialist BI skills. The Data Engineer title alone does not guarantee that capability; responsibilities with the Power BI specialist must be explicit.

9.2.3 Conditions and limits

The target model could retain a shared platform and standards, an embedded or dedicated FinOps Data Engineer for data products, and Power BI collaboration for reporting. Greater business proximity is not evidence of a Data Mesh implementation.

Changes remain subject to access controls, tests, business acceptance and publication rules. Peer review and written handover reduce dependence on one person.

An embedded role adds cost and may not justify a full-time position given the request volume. Compare a part-time dedicated resource and a mixed model as well. External delivery may remain valuable for specialized or occasional needs.

9.2.4 Proposed evaluation

For a small set of representative authorized requests, record request, approval, handoff, work start, delivery and acceptance dates, plus active time by role and iteration count. Separate elapsed delay from human effort.

Compare observations with a clearly defined embedded scenario, or with a measured pilot if one exists. A simulation does not establish actual company improvement. Chapter 8 will value effort, external services and recurring costs without assuming rates or savings.